Ashish Gupta

Bachelor of Technology
in Computer Science Engineering
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GitHub | Linkedin
Codeforces | Codechef

Education

Degree	Institute/Board	CGPA/Percentage	Year
Bachelor of Technology	Indian Institute of Technology, Ropar	8.19 (Till 4th Sem)	2022-2026
Senior Secondary	M.R.S Public School (CBSE)	93.4%	2021
Secondary	Central Academy Schools (CBSE)	98%	2019

Experience

- **ML Internship - IEEE SPS VIP Cup: Retinal OCT Image Classification** *April 2024 - June 2024*
  ML intern under the supervision of Dr.Puneet Goyal, IIT Ropar **Github**
  - Incorporated deep learning methods to classify noisy retinal OCT images into Healthy, DME, and Non-Diabetic categories. The team was chosen as one of the three finalists globally to present at the ICIP Conference in Abu Dhabi.
  - **Techniques adopted**
    - ***Denoising task***: Employed a Self-Supervised Self2Self Network (encoder-decoder architecture) and Conditional GAN (cGAN) to enhance image quality by reducing noise up to 90%.
    - ***Super Resolution***: Engineered 2Encoder-1Decoder ESRGAN and Conditional GAN (cGAN) models to improve the clarity of low-resolution images in the test dataset, enhancing the quality and detail of retinal OCT images.
    - ***Classification***: Leveraged pre-trained ResNet-50 combined with the Channel Block Attention Module (CBAM) for accurate volume classification post-denoising and super-resolution.
  - **Models implemented:** Conditional GAN (cGAN), Self2Self Network, 2Encoder-1Decoder ESRGAN, Through Plane SR cGAN, ResNet-50 with CBAM.

Projects

- **Marker-Based Image Segmentation using Disjoint Set Union**
  Link | *Python | Flask | HTML | CSS | JavaScript | Data Structures | DSU* **GitHub**
  - Coded Disjoint Set Union for efficient image segmentation, isolating objects using user-provided markers.
  - Coordinated Sobel operators for gradient image conversion and marker extraction to enhance region identification, achieving segmentation results with 95% accuracy. Achieved time complexity of $O(\alpha(n))$, where $\alpha(n)$ is the inverse Ackermann function that grows extraordinarily slowly. Also, deployed a *full-stack responsive website* using Flask for the backend.
- **Game of Cops and Robbers on Graphs**
  Visual Studio | C++ | SFML Library of C++ **GitHub**
  - Crafted a game simulating Cops and Robbers scenarios on graph topologies such as grids, cylinders, and torus (sphere shaped) graphs, ensuring the algorithm accomplishes the task in the minimum possible time (80% efficient) and minimal number of cops.
  - Integrated a responsive Graphical User Interface (GUI) using SFML. Enabled player interaction through mouse clicks for cop placement and keyboard inputs (normal 4 arrow keys) for controlling robber movement.
- **Minor Projects | GitHub**
  - **Reducing Cache Conflict Misses with ChampSim:** Adopted strategies to reduce conflict misses at the Last Level Cache by analyzing and categorizing cache sets based on access and eviction counts into Very Hot, Hot, Cold, and Very Cold. Optimizations implemented led to a 7% reduction in cache misses, enhancing overall system efficiency and throughput, and placing the team in the top 10% of performers.
  - **RISC-V Assembler:** Facilitated a 32-bit RISC-V assembler similar to Venus, converting RISC-V instructions to assembly language and calculating branch prediction results up to 2-bit accuracy.
  - **CUCU Compiler:** Devised a compiler using Lex and Yacc, featuring a lexer and parser for a toy programming language.
  - **Random Walk Algorithm:** Executed a randomized algorithm to determine the leader from a group of people.
  - **Songify-Music Player:** Developed a Front-end music player using HTML, CSS, and JavaScript, featuring play/pause functionality, and the ability to toggle to next/previous tracks.

Technical Skills

- **Programming Languages**: C/C++, Python, Tensorflow, Keras, NumPy, HTML, CSS.
- **Competitive Programming**: C++, Codeforces: **1424** Specialist, CodeChef: **1846** 4 Star.
- **Softwares/Tools/Libraries**: Git, Github, VS Code, Arduino IDE, Unix/Linux (Ubuntu).
- **Key Courses Taken**: Computer Architecture, DSA, Discrete Mathematics, Programming Paradigms.

Achievements

- **VIP Cup**,Selected among the 3 finalist teams globally to present the finals at the ICIP Conference in Abu Dhabi. *2024*
- **Codechef**,Achieved a global rank of 38 in Starters 141 and 142 in Starters 136. *2024*
- **Highest CGPA of 9.78**,in the 1st semester. *2023*
- **Top 1% of academic batch**,Awarded Institute Merit Scholarship for being in the top 1% of the batch. *2023*
- **Branch Change Award**, Sole branch changer from MnC to CSE with a CGPA of 9.32. *2023*
- **Ranked**, in the top 1% of candidates in **JEE-Advanced** out of over 250,000 candidates across India. *2022*
- **City Rank 2 in Secondary Examinations**, 98% marks obtained. *2019*